



NRC MEETING

April 11, 2024

**9 am to 11 am, Pinckney Hall
Ballroom**



**BEAUFORT-
JASPER
WATER & SEWER
AUTHORITY**

Sun City Neighborhood
Representative Council

April 11, 2024

DISCUSSION PLAN

- I. BJWSA overview
 - A. Water treatment / infrastructure
 - B. Wastewater treatment / infrastructure
- II. Our ongoing challenges
- III. Our plans to address those challenges
 - A. How you can help
- IV. Q & A / Discussion



Water Treatment Facts

Water Treatment is one of the greatest factors in extending the human lifespan
(Antibiotics get an honorable mention)

The first Community in the U.S. began disinfecting water in 1908

- Thousands soon followed (decades before antibiotics)
- Life threatening cholera and typhoid diagnosis dropped dramatically
 - Typhoid fever occurrence in the US dropped from 100/100,000 to .1/100,000
 - 75% of that .1 occurring among international travelers

Water treatment is a factor that distinguishes “developed” countries



MISSION

Provide quality water and wastewater services to our current and future customers in the Lowcountry.

2

COUNTIES

Beaufort
Jasper

4

**MUNICIPALITIES
SERVED**

Bluffton
Beaufort
Hardeeville
Port Royal

48,400

**SEWER
Connections**

750

SQUARE MILES

Approximate
Service Area

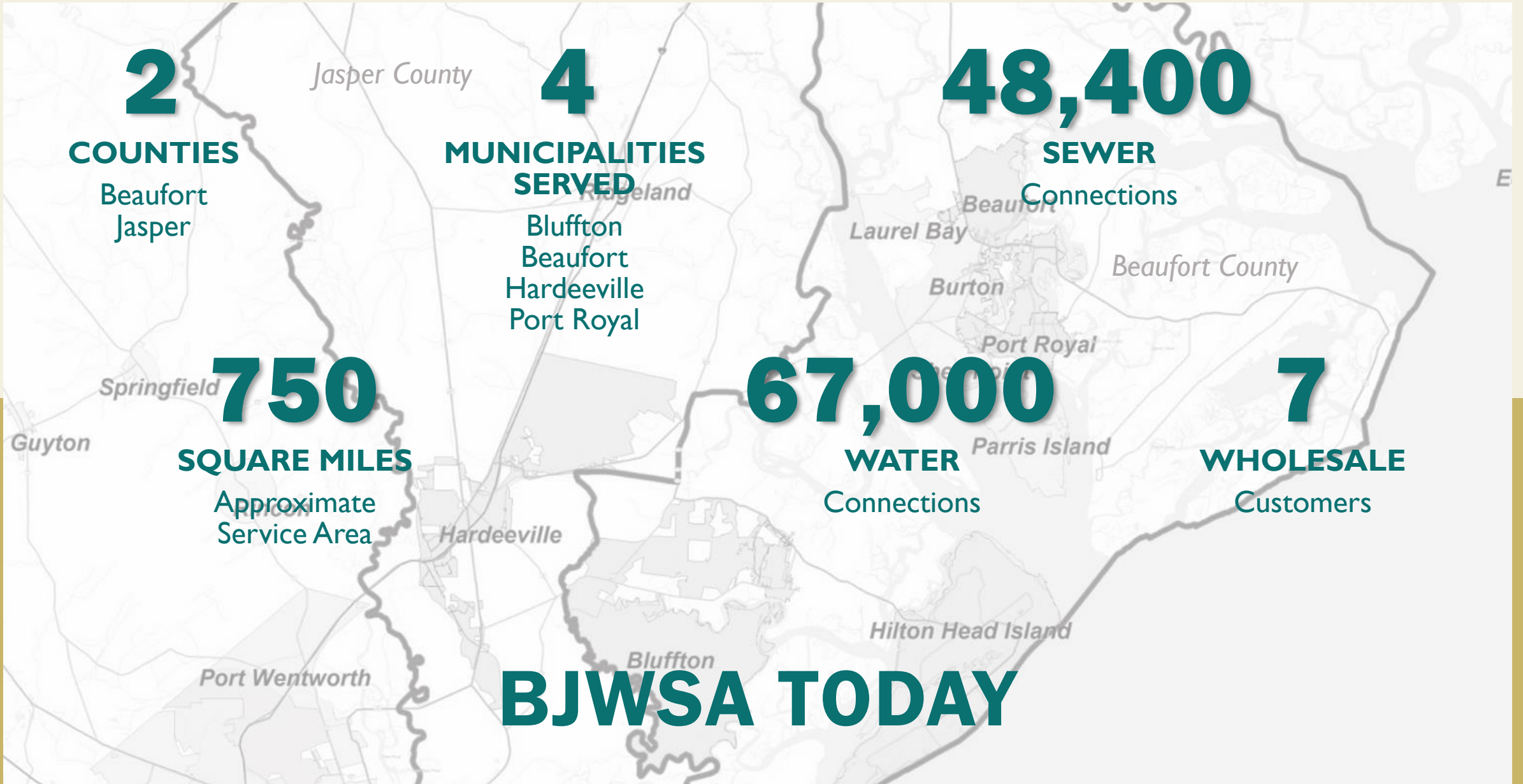
67,000

**WATER
Connections**

7

**WHOLESALE
Customers**

BJWSA TODAY



WATER SYSTEM

SUMMARY

Primary Water Source

- Savannah River
- 18-mile long canal

Total treatment capacity: 39 mgd

- Chelsea WTP, 24 mgd
- Purrysburg WTP, 15 mgd

Distribution

- Greater than 1,500 miles of water main
- 10 booster pump stations
- 8,701 fire hydrants
- 25,798 valves
- 6 mg stored in Elevated Tanks for peak use
- Aquifer Storage and Recovery (ASR) underground storage



RAW WATER CANAL

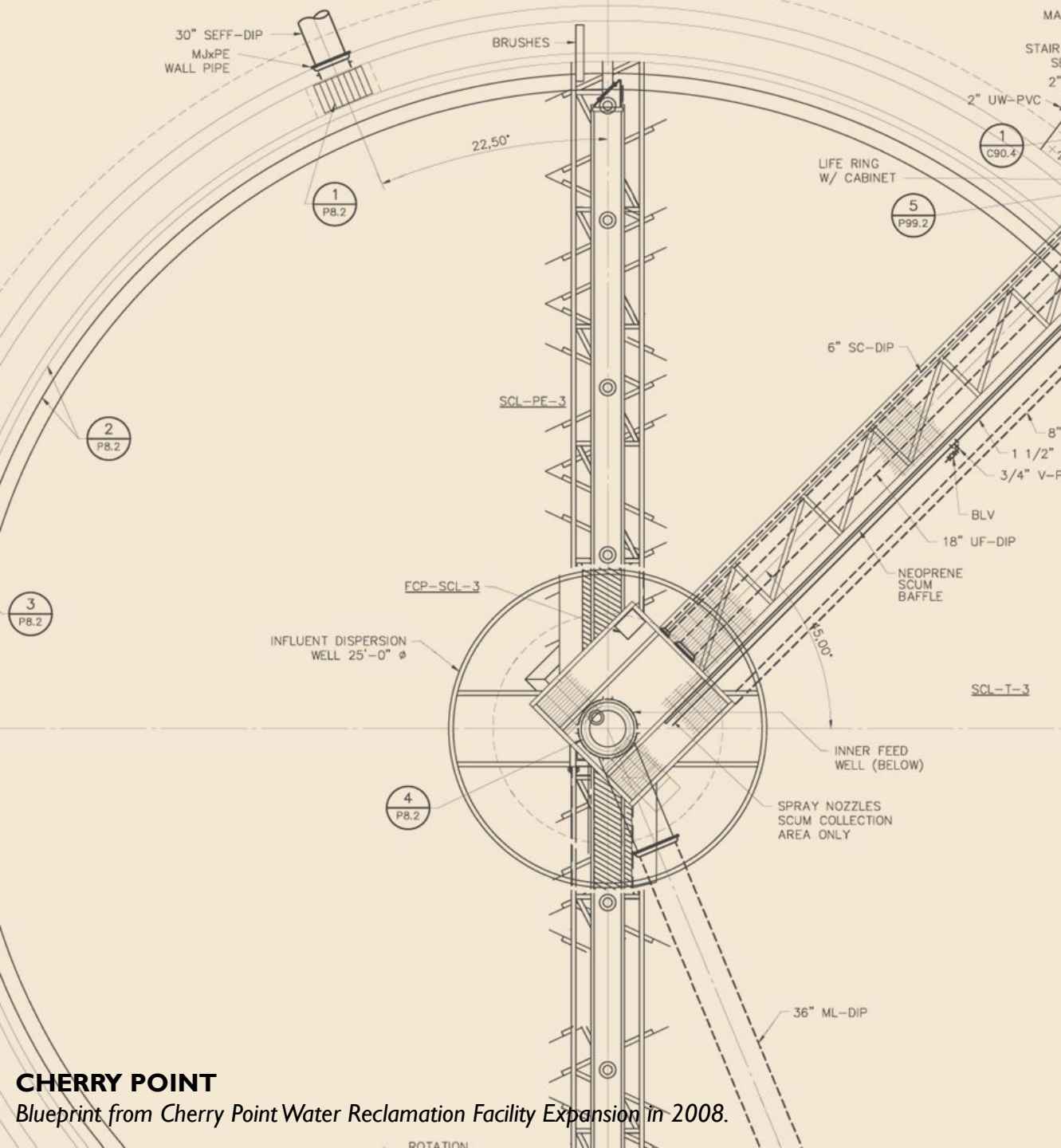
This 18-mile canal carries raw water from the Savannah River for treatment.

WATER SYSTEM

Our Shared
Watershed



Savannah River Basin



WASTEWATER TREATMENT

SUMMARY

Total Treatment Capacity

- Average of 10.5+ mgd treated
- 8 treatment plants with total capacity of 20.2 mgd

Collection System

- Serves approx. 48,400 retail customers
- 1,000+ miles of collection main
- 600+ lift stations

Treatment

- Biological treatment, UV light, and chemical disinfection

Sustainability

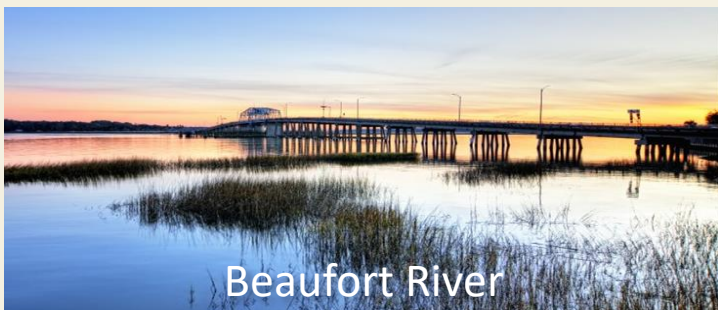
- Solar array generates 1,200 Mwh of energy annually

CHERRY POINT

Blueprint from Cherry Point Water Reclamation Facility Expansion in 2008.

Discharge: *Where does it go?*

Once treated to near drinking water standards water is discharged to:



Great Swamp Effluent Management System helps to protect 980 acres of natural forested wetlands.

Golf Course and select neighborhood irrigation (March to September).

Savannah, Broad and Beaufort Rivers

BJWSA Fiscal Sustainability:

- Meeting existing and future requirements for health and safety.
- Balancing rates with needs for reliability, safety and environmental responsibility.
- Investing to meet customer demands.
- Adjusting to higher costs associated with staffing, healthcare, chemicals, energy, and materials as well as supply chain issues.





STRATEGIC PLAN



INVEST IN OUR WORKFORCE

Through recruiting, retaining, developing, and training our workforce.



PROMOTE FINANCIAL SUSTAINABILITY

Through maintaining a responsible rate structure and assuring financial compliance.



MANAGE RISK & LEVERAGE INNOVATION

Through providing a safe work environment, practicing sustainability and stewardship, and improving emergency response.



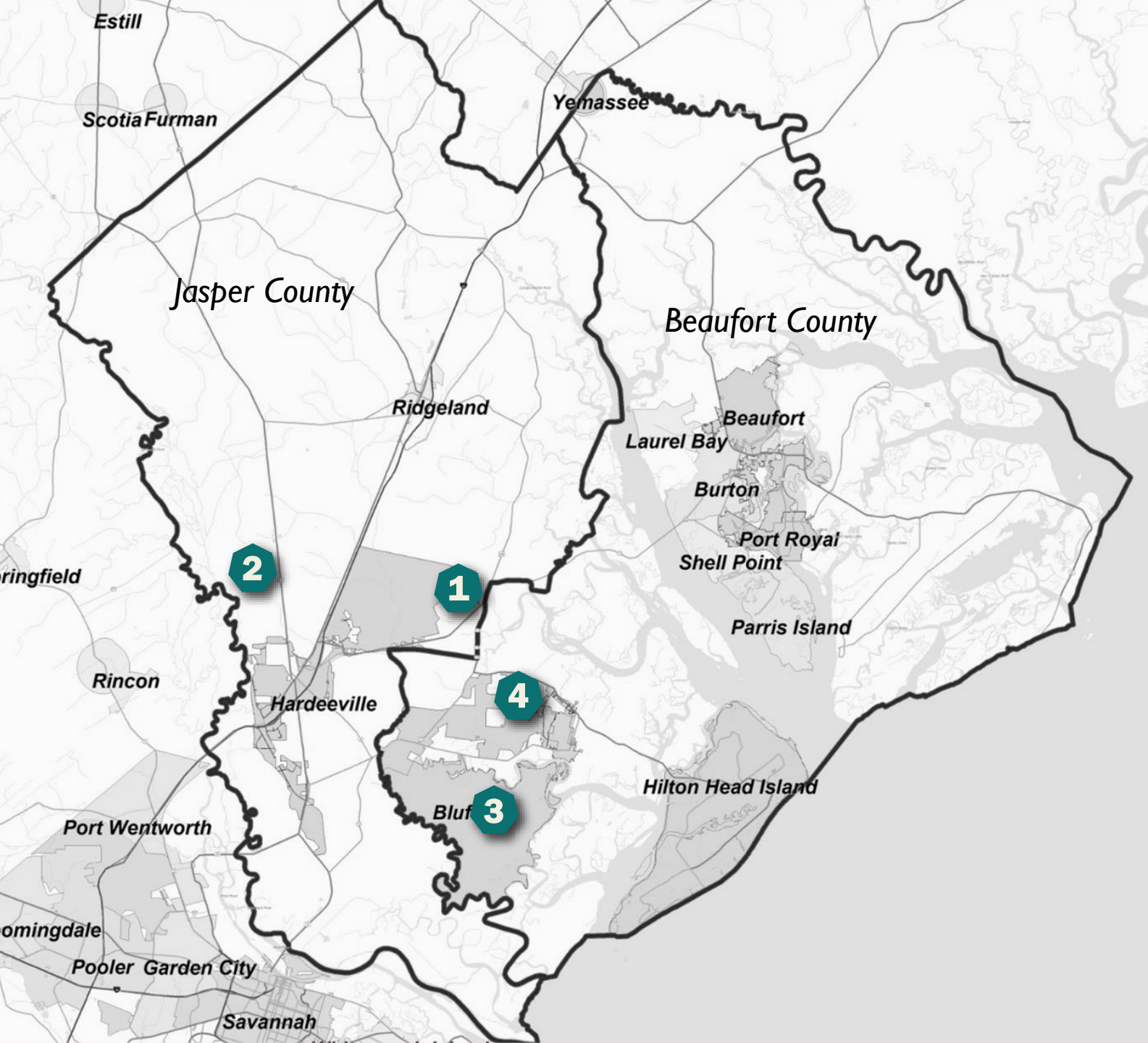
ENSURE INFRASTRUCTURE SYSTEM VIABILITY

Through master planning, leveraging asset management and technology, and optimizing system performance.



ENGAGE CUSTOMERS AND STAKEHOLDERS

Through enhancing customer education and strengthening municipal and stakeholder coordination.



1 CHERRY POINT
WASTEWATER
TREATMENT
PLANT

Expansion from
7.5 mgd to 11.25 mgd

3 MAY RIVER
WATERSHED
PROTECTION

Stoney Creek
Sewer Project

2 PURRYSBURG
WATER
TREATMENT
PLANT

Expansion from
15 mgd to 30 mgd

4 BLUFFTON
STORAGE TANK

1.5 million gallon
Elevated Storage Tank
Construction

KEY PROJECTS

OUR CHALLENGES

1 **Service demand**
~3,000 new connections/year

2 **Cost increases**

Construction materials, equipment and chemicals have increased between 20-50% over the last few years

3 **Regulatory changes**

Federal and state regulations require new levels of treatment and limit abilities to discharge

4 **Construction limitations**

It takes significantly longer to plan, permit, and build a wastewater treatment plant than it does to build residential units

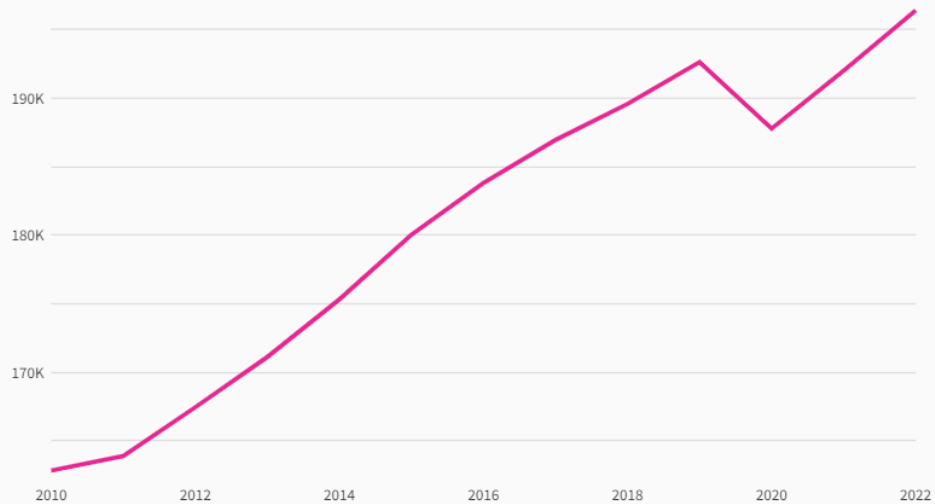
Supply chain issues still exist and historic national infrastructure spending is putting a heavy demand on US contracting capacity.



How many people live in Beaufort County?

Beaufort County's population **grew 20.6%** from the **162,843** people who lived there in **2010**. For comparison, the population in the US **grew 7.7%** and the population in South Carolina **grew 14%** during that period.

Population in Beaufort County



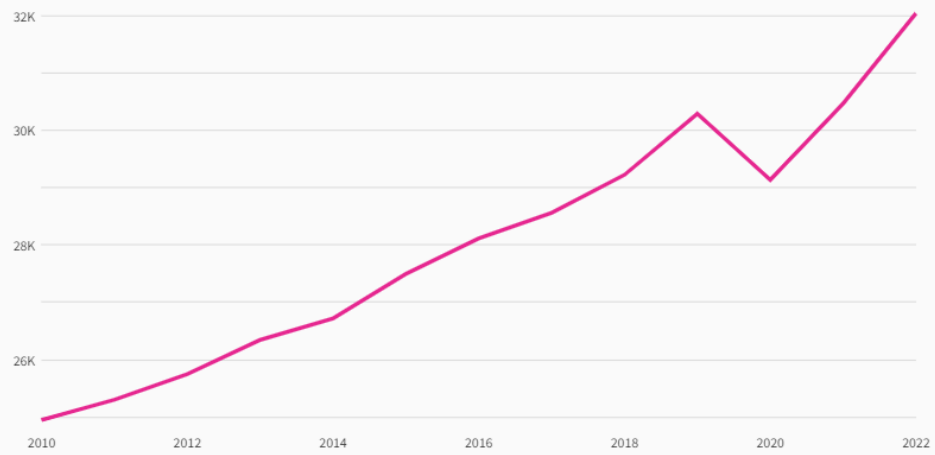
Population 2010 – 2022

- **7.7%** United States
- **14%** South Carolina
- **20.6%** Beaufort County
- **28.4%** Jasper County

How many people live in Jasper County?

Jasper County's population **grew 28.4%** from the **24,949** people who lived there in **2010**. For comparison, the population in the US **grew 7.7%** and the population in South Carolina **grew 14%** during that period.

Population in Jasper County



AREA GROWTH
TOTAL POPULATION

WHAT IS BJWSA DOING TO MEET MARKET CHALLENGES?



1 Expediting Investment

Seeking regulatory approval and building at a faster rate

- Wastewater treatment
- Wastewater collection and pumping
- Drinking water treatment and storage
- Drinking water storage

2 Creative Solutions

- **Diverting wastewater** from Cherry Point to other treatment plants
- Exploring options **expedite flow through the Great Swamp** to improve the natural capacity
- Working with landscapers and property owners to **mitigate morning water usage peaks**

3 Recruiting New Staff

Hiring new professionals to meet the needs of our ever increasing customer base and infrastructure needs

OUR REALITIES

- **Traditional infrastructure planning timelines are being outpaced** by an unprecedented rate of growth
- **Utility construction is much more expensive** than it was just five years ago and challenged by supply and contracting availability
- **Unfunded regulatory mandates** with extremely short compliance timelines are further stressing finances
- **Fees for capacity (Developers) need to substantially increase**
- **Rates need to keep pace** with operating and regulatory cost increases



HOW YOU CAN HELP

(AS A COMMUNITY LEADER)

**Understand that
unprecedented growth is
straining our infrastructure-
Collaborative planning is critical**

Help educate residents about
system challenges and the need to
follow our Irrigation Management
Plan



HOW YOU CAN HELP

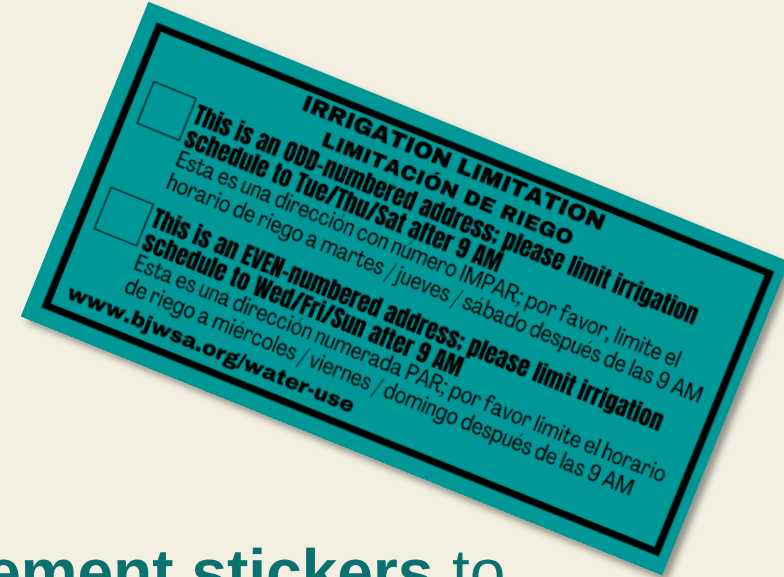
(AS AN INDIVIDUAL)

Adjust irrigation controls to water no more than three days per week.

- Tuesday, Thursday and Saturday for odd-numbered addresses
- Wednesday, Friday and Sunday for even-numbered addresses

Avoid irrigating on Mondays and between the hours of 3 AM and 9 AM when system demand is the most extreme

Apply Irrigation Management stickers to control panels and mark the appropriate box that corresponds with the address.



HOW YOU CAN HELP

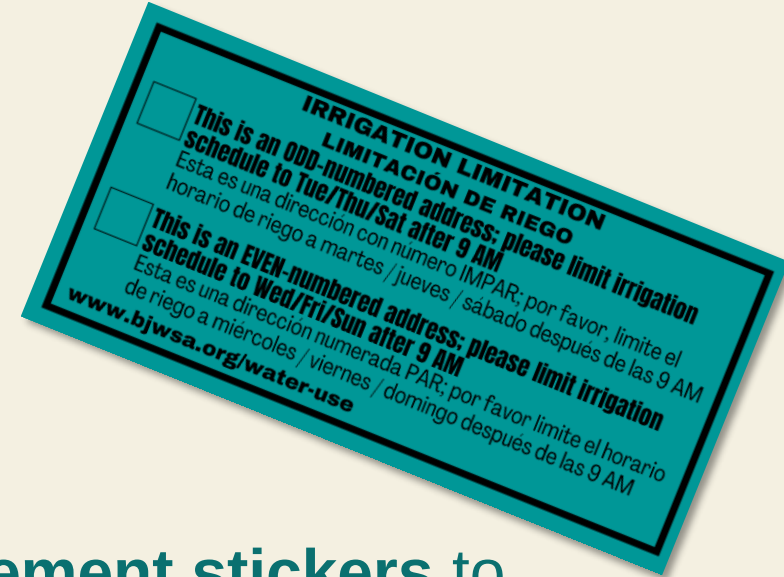
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THANK YOU



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